EDITH Next-Gen AML Intelligence Platform

End-to-End Engineering Workplan & Execution Ledger (v2.0)

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Version: 2.0
Status: Phases 1–5 Completed & Verified (93/93 Automated Tests Passing)
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1. Phase Execution Summary

Phase	Module / Deliverable	Key Engineering Milestones	Status
Phase 1	**Local Infrastructure & Dual-Database**	Docker Compose stack (`PostgreSQL 17` on `:5432` + `OpenSearch 2.12` on `:9200`), 5-schema DDL migrations (`core` , `screening` , `cases` , `rules` , `notifications`), reference seeds.	**COMPLETED**
Phase 2	**OpenSearch 5-Pass Screening Engine**	`Edith.Screening.Core` & `Edith.Screening.Api` , Double Metaphone phonetic analyzer, n-gram DisMax query, Jaro-Winkler + Damerau-Levenshtein scoring, idempotency store, whitelist bypass.	**COMPLETED**
Phase 3	**Bulk Streaming Ingestion & Deduplication**	`Edith.Screening.Bulk` , streaming `CSV` / `XML` / `JSON` parser for 9 AML fields, duplicate entity skip detection, `keepSkipped` retention, and on-demand export in `CSV` / `XML` / `JSON` .	**COMPLETED**
Phase 4	**Case Investigation & 4-Eyes Governance**	`Edith.Cases` , automatic case creation from `HELD` transactions, 15-min pessimistic lease locking, multi-type comments, 4-Eyes Maker-Checker RBAC, HMAC-SHA256 CBS webhooks (`RELEASE` / `FREEZE`), `goAML` XML & AI Narrative Copilot.	**COMPLETED**
Phase 5	**Frontend Live Integration & UI Bridge**	Connected React 19 `edith-portal` (`:3000`) to `Edith.Gateway` (`:5000`), real JWT persona authentication, live case workspace escalation/resolution/comments, and live bulk file upload + skipped download in UI.	**COMPLETED**

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2. End-to-End Operational Data Flow

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flowchart LR
    A["Core Banking / Portal Upload"] --> B["Edith.Gateway (:5000)"]
    B --> C["5-Pass Screening Engine (OpenSearch 2.12)"]
    C -->|Score < 80% or Whitelisted| D["CLEARED"]
    C -->|Score >= 80%| E["HELD + Auto-Create Case"]
    E --> F["Analyst L2 Triage (15m Lease Lock)"]
    F -->|Escalate| G["PendingApproval (MLRO Queue)"]
    G -->|False Positive| H["CBS Webhook: RELEASE"]
    G -->|True Hit| I["CBS Webhook: FREEZE + goAML STR XML"]
```

3. Verification & Quality Gates

- **Solution Build:** `dotnet build Edith.slnx` — 0 warnings, 0 errors.
- **Automated Test Suite:** `dotnet test Edith.slnx` — **93/93 tests passing** across 10 test projects.
- **Frontend Production Bundle:** `npm run build` in `src/Web/edith-portal` — `tsc -b && vite build` clean with 0 errors.